Title: Exploratory Data Analysis and Data Cleaning: Titanic Dataset

1. **Dataset Overview:** The Titanic dataset has 891 passengers and 12 columns (passenger details such as class, sex, age, fare, and whether they survived). Source: Kaggle.
2. **Data quality:** Age was missing in 177 rows (19.9%), Cabin in 687 (77.1%), and Embarked in 2 (0.2%). There were no duplicate rows.
3. **Cleaning methodology:** Age was filled with the median (28), Embarked with the mode (S), and Cabin was dropped. Fare outliers were capped using the IQR method (upper limit about 65.6). Survived and Pclass were converted to categorical types. The cleaned data has 891 rows and 11 columns.
4. **Key insights:** your 3 observations, plus the survival gap between women and men.
5. **Challenges faced:**
    - Choosing how to treat missing data
    - Deciding between removing and capping outliers
    - Working across phone and laptop
    - Fixing a “file not found” error and a missing-brackets bug
    - Doing the IQR method for the first time so it was difficult to understand.